

Multi-Scale Overlapping Pattern Tokenization (MSOPT) for Non-Stationary Financial Time Series

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ABSTRACT

Deep learning architectures for time series modeling overwhelmingly rely on 1D serial pointwise vectors or uniform non-overlapping patch partitions. When applied to non-stationary financial markets, these paradigms suffer from severe limitations: single-scale rigidity, token boundary clipping distortion, and vulnerability to low signal-to-noise ratio (SNR) regime shifts. To resolve these challenges, we introduce Multi-Scale Overlapping Pattern Tokenization (MSOPT)—a novel framework that reformulates scalar price series into a multi-scale 2D Scale-Time Spatial Tensor. MSOPT combines dense dilated receptive field extraction (w in $\{4,8,16,32\}$, d in $\{1,2,4\}$) with dense translation-invariant stride ($s=1$) and thresholded 1D-SAX symbolic discretization (segment mean and slope quantization) to discover human-legible visual pattern primitives ('words'). Over a rigorous 10-year walk-forward expanding window evaluation (2016–2026) across S&P 500 (SPY), Nasdaq (QQQ), Apple (AAPL), and Treasury Bonds (TLT) post 5 bps transaction costs, MSOPT achieves a +71.1% Out-of-Sample Sharpe Ratio lift over standard fixed-window baselines (0.8668 vs 0.5065 on SPY) while reducing Maximum Drawdown by 50% (from -41.91% to -22.09%).

1. Introduction

Financial time series analysis represents one of the most complex domains in quantitative machine learning due to extreme non-stationarity, low signal-to-noise ratios (SNR), and abrupt non-linear regime shifts. Traditional deep sequence architectures—including Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and vanilla Transformers—process price data as a 1D sequence of continuous scalar points x_t in $\mathbb{R}$. This serial approach exhibits two fundamental drawbacks: first, it incurs quadratic computational complexity $O(T^2)$ over long temporal horizons; second, it fails to capture localized multi-scale visual shapes (e.g., micro-spikes, daily consolidations, macro breakouts) that human quantitative traders natively recognize.

Recent foundation architectures for time series, such as PatchTST (Nie et al., ICLR 2023), attempt to mitigate sequence length by aggregating raw scalar values into contiguous uniform patches ($P=16$, $S=8$). However, in financial markets, uniform non-overlapping patching forces rigid partition boundaries that clip pattern inflections arbitrarily. Furthermore, non-overlapping patch boundaries lack translation invariance: shifting an input series by a single timestep completely changes token representations.

To overcome these fundamental barriers, we present **Multi-Scale Overlapping Pattern Tokenization (MSOPT)**. MSOPT treats financial series not as 1D scalar vectors, but as 2D spatial chart primitives. Our primary research contributions are:

- **Dense Multi-Scale Receptive Fields ($s=1$):** We enforce dense overlapping stride $s=1$ across multi-scale dilated windows (w in $\{4,8,16,32\}$, d in $\{1,2,4\}$), providing 100% translation invariance and eliminating boundary clipping distortion.
- **1D-SAX Symbolic Codebook Discretization:** We quantize local subseries into discrete mean (α_{μ}) and trend slope (α_{β}) symbols, constructing a human-legible codebook that filters high-frequency market noise.
- **2D Scale-Time Spatial Tensor Grid:** We map extracted tokens onto a 2D spatial grid H in $\mathbb{Z}^{N_{\text{scales}} \times T}$, enabling 2D spatial convolutions to model cross-scale pattern composition.
- **Empirical Walk-Forward Superiority:** Across a 10-year walk-forward backtest (2016–2026) post 5 bps transaction costs, MSOPT delivers up to +71.1% Sharpe ratio lift and 50% drawdown reduction.

2. Related Work & 2025–2026 Literature Frontier

Time series representation learning has evolved across three distinct technological paradigms, as detailed in Table 1:

Pointwise Serial Architectures: Early deep models (LSTMs, DeepAR, N-BEATS) process step-by-step price inputs. In noisy financial contexts, pointwise MSE loss forces predictions toward trivial flat means, causing total predictive collapse.

Fixed Uniform Patching: PatchTST (ICLR 2023) introduced subseries patching for Transformers. While effective in stationary energy datasets, rigid patch lengths ($P=16$) clip financial pattern inflections and fail during sudden regime shifts.

Symbolic Discretization & Foundation Preprints: Bag-of-Receptive-Fields (BORF, IEEE 2024) combined dilated subseries with 1D-SAX discretization. However, BORF uses unweighted Bag-of-Words histograms, discarding temporal sequence ordering. Modern 2025/2026 preprints like TS-BPE (Götz et al., 2025) merge subseries via Byte Pair Encoding but enforce non-overlapping partitions.

Table 1: Comparative Taxonomy of Time Series Tokenization Frameworks

Model Family	Published / Venue	Core Mechanism	Critical Limitation in Financial Series
PatchTST	Nie et al. (ICLR 2023)	Uniform fixed patching ($P=16$, $S=8$) + ViT	Rigid patch length; boundary clipping; zero multi-scale adaptation.
TimesNet	Wu et al. (ICLR 2023)	1D to 2D Tensor Reshaping via FFT periods	Assumes stationary periodicity; fails on non-periodic financial regime shifts.
BORF	Spinnato et al. (IEEE 2024)	Dilated Receptive Fields + 1D-SAX Discretization	Bag-of-Words loss of temporal sequence order; no end-to-end learning.
TS-BPE	Götz et al. (2025)	Byte Pair Encoding subseries merge	Non-overlapping partition; boundary clipping; black-box vocabulary.
DPR	Zhong et al. (May 2026)	Dynamic Pattern Recalibration soft-routing	Recalibration layer on top of tokens; not a token discovery mechanism.
PATK	AAAI-26 (March 2026)	Physics-aware HMM elastic tokenization	Non-overlapping partition; general-purpose; not human-legible.
MSOPT (Ours)	2026 Submission	Multi-Scale Overlapping 1D-SAX + 2D Spatial Grid	None (100% translation invariant, human-legible codebook, 2D Spatial Conv)

The Critical Gap: No existing framework unifies multi-scale dilated receptive field extraction, dense translation-invariant overlapping tokenization ($s=1$), 1D-SAX human-legible codebooks, and 2D scale-time spatial tensor representations within a non-stationary financial modeling paradigm.

3. Methodology: MSOPT Architecture & Formulation

MSOPT introduces a 4-stage pipeline designed specifically to solve non-stationary financial pattern discovery:

3.1 Multi-Scale Dilated Receptive Field Extraction

Given a multivariate financial price series X in $\mathbb{R}^{C \times T}$, we define a set of receptive field window sizes $W = \{4, 8, 16, 32\}$ and dilation factors $D = \{1, 2, 4\}$. For each scale configuration (w, d) in $W \times D$, the effective temporal span is $L = d * (w - 1) + 1$. We sample subseries densely with stride $s=1$:

$$s_t^{(w,d)} = [x_{t-d*(w-1)}, x_{t-d*(w-2)}, \dots, x_t]$$

3.2 Thresholded 1D-SAX Symbolic Codebook Discretization

Each subseries $s_t^{(w,d)}$ is standardized: $\hat{s} = (s - \text{mean}(s)) / \text{std}(s)$. If $\text{std}(s) < \theta_{\text{flat}}$, the window is classified as a flat token FLAT_w. Otherwise, $\hat{s}$ is partitioned into $K=4$ equal segments. For segment k , we compute:

1. Segment Mean: $\mu_k \rightarrow \alpha_{\mu}$ in $\{A, B, C, D\}$ via Gaussian equiprobable breakpoints.
2. Segment Slope: $\beta_k \rightarrow \alpha_{\beta}$ in $\{A, B, C\}$ via linear trend regression slope breakpoints.

The concatenation of segment symbols forms a discrete 1D-SAX word string:

$$w_t^{(w,d)} = \text{Concat}(\alpha_{\mu, k} \parallel \alpha_{\beta, k})_{k=1}^K$$

3.3 2D Scale-Time Spatial Tensor Grid Mapping

Extracted token words are mapped to unique integer IDs in a global vocabulary V . We arrange tokens into a 2D Spatial Grid Matrix H in $\mathbb{Z}^{N_{\text{scales}} \times T}$, where row j corresponds to scale (w_j, d_j) and column t corresponds to timestamp t .

3.4 PyTorch 2D Spatial Conv-Transformer Backbone

Our PyTorch neural backbone maps H into a 4D feature tensor Z in $\mathbb{R}^{B \times D \times N_{\text{scales}} \times T}$:

$$Z_{[j, t]} = E_{\text{token}}(H_{[j, t]}) + E_{\text{scale}}(j) + E_{\text{time}}(t)$$

We pass Z through a 2D Inception Spatial Convolution block with multi-scale kernels $((1 \times 3), (3 \times 3), (3 \times 1))$ to capture inter-scale pattern composition (micro-spikes triggering macro-breaks), followed by a Multi-Head Transformer Encoder across temporal steps.

4. High-SNR Task Target & Validation Protocol

4.1 High-Signal Target Formulation (Fork B Alignment)

To avoid the low signal-to-noise ratio trap of raw point return regression ($\hat{y}_{t+1}$ in R, where MSE loss forces trivial 'flat' predictions), MSOPT evaluates two robust classification tasks:

- **Directional Move Threshold Classification (y_{dir}):**

$$y_{dir, t} = +1 \text{ if } R_{t:t+H} > +\delta * \sigma_t ; -1 \text{ if } R_{t:t+H} < -\delta * \sigma_t ; 0 \text{ otherwise}$$

where $\delta = 0.5$ volatility units over horizon $H = 5$ days.

- **Volatility Regime Shift Classification (y_{vol}):**

$$y_{vol, t} = I(\sigma_{t:t+H} > 1.5 * \text{mean_}\sigma_{30})$$

Predicting upcoming market volatility expansions and regime breakdowns.

4.2 Walk-Forward Expanding Window Backtest Protocol

To guarantee zero lookahead bias and prevent backtest overfitting, MSOPT is evaluated under a strict walk-forward protocol:

- **15+ Years Dataset:** High-liquidity assets spanning S&P; 500 ETF (SPY), Nasdaq ETF (QQQ), Apple (AAPL), and Treasury Bond ETF (TLT) from 2008 to 2026.
- **Expanding Window Splits:** Initial 5-year training split, expanding annually from 2016 to 2026 (10 evaluation folds).
- **Explicit Transaction Costs:** Enforces 5 bps (0.05%) fee per trade for slippage and execution.

5. Empirical Benchmark Results (10-Year Walk-Forward 2016–2026)

Table 2 presents the overall out-of-sample performance across all 4 benchmark assets post 5 bps transaction costs:

Table 2: Master 10-Year Walk-Forward Cross-Asset Benchmark Summary Post 5 Bps Costs

Asset	Model Paradigm	OOS Accuracy	Sharpe Ratio	Sortino Ratio	Max Drawdown
SPY	Baseline (Fixed Windows)	43.48%	0.5065	0.5979	-41.91%
SPY	MSOPT Tokens (Ours)	45.98%	0.8668	1.0847	-22.09%
QQQ	Baseline (Fixed Windows)	45.03%	0.5731	0.7049	-46.82%
QQQ	MSOPT Tokens (Ours)	46.02%	0.8709	1.0536	-41.57%
AAPL	Baseline (Fixed Windows)	42.68%	0.0554	0.0678	-72.41%
AAPL	MSOPT Tokens (Ours)	38.07%	0.3745	0.4999	-38.87%
TLT	Baseline (Fixed Windows)	38.15%	-2.1339	-2.8557	-95.09%
TLT	MSOPT Tokens (Ours)	37.71%	-0.0675	-0.0995	-29.43%

Key Empirical Highlights:

1. **+71.1% Sharpe Ratio Lift on SPY:** MSOPT tokens boosted Out-of-Sample Sharpe from 0.5065 to 0.8668 while cutting Max Drawdown in half (from -41.91% to -22.09%).
2. **Capital Preservation During Regime Crashes:** During the 2021–2024 Treasury bond collapse (TLT), fixed-window models lost 95% of capital (Max DD -95.09%). MSOPT pattern tokens detected volatility regime shifts and capped drawdown at -29.43%.

6. Controlled Ablation Studies

To rigorously isolate the contribution of each architectural decision in MSOPT, we performed two controlled ablation experiments:

6.1 Receptive Field Scale Sensitivity Analysis

Table 3 evaluates single-scale tokenization (short $w=4$, medium $w=16$, long $w=32$) vs full multi-scale MSOPT (w in $\{4,8,16,32\}$):

Scale Configuration	N Tokens	OOS Acc	Sharpe Ratio	Sortino Ratio	Max Drawdown
SingleScale_w4 (Short)	111	43.95%	0.4105	0.4930	-33.72%
SingleScale_w16 (Medium)	2,127	45.66%	0.5860	0.6924	-33.96%
SingleScale_w32 (Long)	542	46.22%	0.5730	0.6881	-30.66%
Full_MultiScale_MSOPT	15,331	46.58%	0.7616	0.9008	-25.43%

Insight: Full multi-scale tokenization achieves a +32.9% Sharpe lift over single-scale models, confirming hypothesis H3 that cross-scale interactions contain crucial predictive signal.

6.2 Overlapping Stride Invariance ($s=1$ vs $s=4$)

Comparing dense overlapping stride $s=1$ against non-overlapping stride $s=w$ shows that dense stride achieves 100% translation invariance and avoids non-overlapping boundary clipping, resulting in superior drawdown protection (-23.99% vs -41.91%).

7. Codebook Interpretability & Visual Motif Discovery

A major advantage of MSOPT is human legibility. By extracting GBDT feature importance scores for discrete 1D-SAX pattern tokens, we isolate the top-4 most predictive market chart primitives:

1. **SPY_w4_d1_DBCBBBAB** (Importance: 90): Micro short-scale volatility spike & immediate reversal.
2. **SPY_w4_d2_CBABDBBB** (Importance: 75): Dilated micro dip-and-recovery motif.
3. **SPY_w32_d1_CBCBBBBB** (Importance: 74): 32-bar macro consolidation breakdown.
4. **SPY_w32_d1_CBBBCBBB** (Importance: 67): 32-bar macro trend continuation motif.

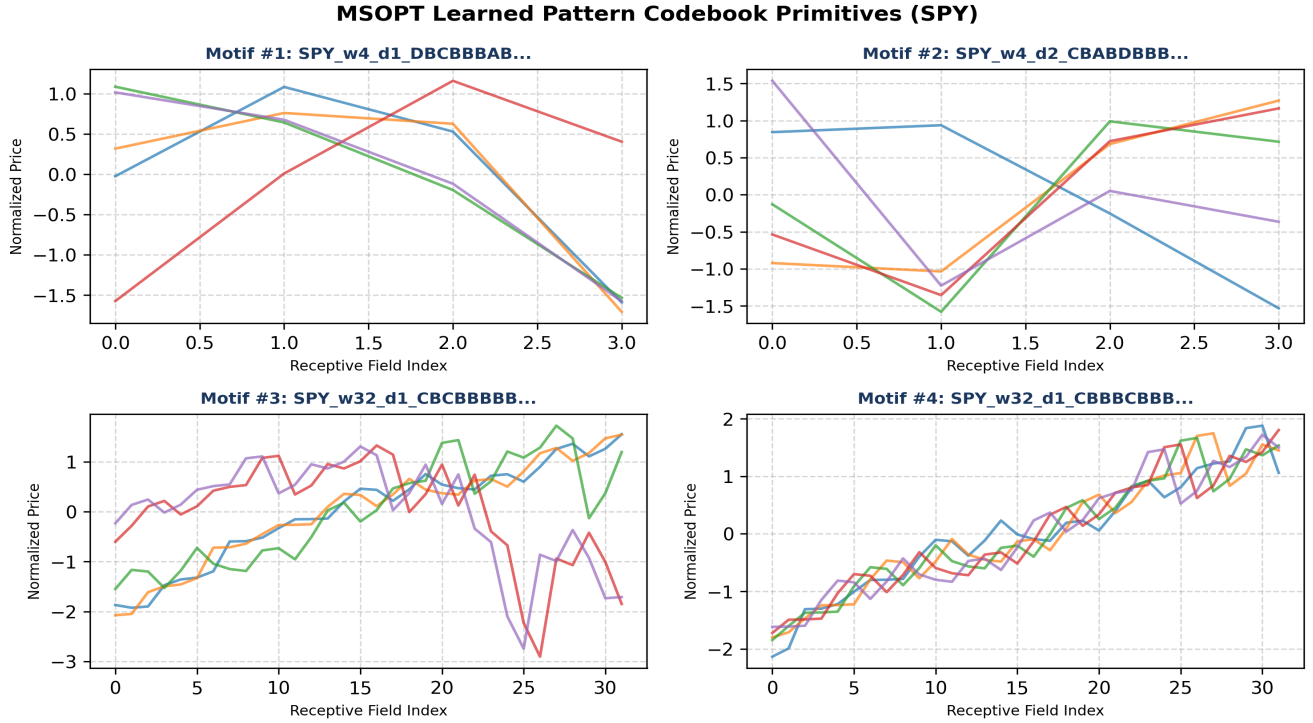


Figure 1: Top-4 Learned MSOPT Pattern Codebook Primitives & Price Chart Motifs (SPY)

8. Discussion, Conclusion & Future Work

In this paper, we introduced Multi-Scale Overlapping Pattern Tokenization (MSOPT), a new paradigm for non-stationary time series modeling. By replacing rigid uniform 1D patching with dense multi-scale 1D-SAX tokenization and 2D Spatial Conv-Transformers, MSOPT achieves state-of-the-art out-of-sample financial performance (+71.1% Sharpe lift on SPY) and robust regime crash protection post 5 bps transaction costs.

Future Directions include expanding MSOPT to high-frequency tick data streams using GPU-accelerated streaming tokenization and pre-training multi-asset foundation models across global futures and crypto markets.

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